Meghana Madhyastha

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SUMMARY

ML systems engineer focused on LLM inference performance: KV-cache management and offloading, prefix caching, multi-tenant GPU serving, model cold-start, and GPU performance on H200, B200 and B300. Works daily in vLLM, SGLang, LMCache and kvcached. PhD in Computer Science from Johns Hopkins, with research on I/O and data-movement bottlenecks in ML, published at ICLR, KDD, ICDE, ICS, HPDC and EuroMLSys.

EXPERIENCE

Parasail · Performance Engineer

Feb 2026 – present

LLM inference provider serving open-weight models on large H200 / B200 / B300 GPU fleets

- **Cold start:** cut engine startup with fastsafetensors + O_DIRECT + io_uring loading and vLLM GPU/CPU snapshot-restore (cuda-checkpoint + CRIU, up to 20× faster startup), plus the InstantTensor loader across vLLM and SGLang (1.5–4.3×). Traced a 293 s → 907 s startup regression to a stale FlashInfer autotuning cache. Blog posts: Making Cold Start Latencies go Brrrr, Restoring vLLM from Snapshots.
- **KV-cache offloading:** took LMCache CPU-memory KV offloading in vLLM to production for a trillion-parameter MLA model, giving a 15× larger prefix-cache pool than GPU memory alone. Extended it to hybrid attention/Mamba models and fixed an MLA KV reload corruption bug.
- **Multi-tenant GPU serving:** built the production layer for kvcached elastic KV-cache GPU sharing, including a Kubernetes device plugin, a node agent, and weight-aware and margin-aware placement. Served 16 models on 5 shared hosts.
- **Measurement:** built a prefix-cache tracing pipeline to measure cross-request reuse, used to size GPU and CPU cache tiers and to rank which models to enable offloading for.
- **Open source:** contributed upstream to LMCache and InstantTensor; maintain patches to vLLM, SGLang and kvcached.

Meta FAIR · Researcher

Dec 2024 – Feb 2026

- Led research on neuron-level activation functions that produce 2:4 and other hardware-friendly sparsity patterns to accelerate LLM pre-training (first author, arXiv 2026).
- Contributed to Composer, a search framework for designing hybrid LLM architectures (ICLR 2026).

Johns Hopkins University · PhD Researcher, Computer Science

2018 – 2024

Advisors: Randal Burns and Joshua Vogelstein

- Designed BLOCKSET (KDD '21) and T-Rex (ICDE '24), memory-layout and traversal techniques that cut inference latency for tree ensembles served from memory and from disk.
- Built DStore (ICS '23) and EvoStore (HPDC '24) with Argonne, model repositories with tensor-level access for storing and serving evolving deep learning models at scale.
- Teaching Assistant for Parallel Programming (CS 320/620), four semesters.

Argonne National Laboratory · Research Intern

Summer 2022

Scalable storage and versioning of deep learning models for HPC workflows (DStore, EvoStore).

Meta · Machine Learning Intern

Summer 2021

Earlier internships

2015 – 2018

CERN (Google Summer of Code, SixTrackLib GPU particle tracking, 2018) · INRIA Parietal lab (2017) · Siemens R&D (2015)

EDUCATION

Johns Hopkins University, PhD, Computer Science — 2018 – 2024

IIIT Bangalore, BS + MS · Master's thesis advisor: Manish Gupta — 2013 – 2018

SELECTED PUBLICATIONS

Y. Shen, **M. Madhyastha**, R. Underwood, B. Nicolae, R. Burns. *Recency/Frequency Adaptive KV Caching for Large Language Model Serving.* ICLR Workshop (AdaptFM) 2026

M. Madhyastha, D. Haziza, J. Cai, N. Ardalani, Z. Bu, C.-J. Wu. *To 2:4 Sparsity and Beyond: Neuron-level Activation Functions to Accelerate LLM Pre-Training.* arXiv 2026

L. Xue, **M. Madhyastha**, M. Lee, A. Storkey, R. Burns, M. K. Marina. *On Harnessing Idle Compute at the Edge for Foundation Model Training.* EuroMLSys 2026

B. Acun, P. Sinha, N. Ardalani, S. Bae, A. Golden, C.-Y. Lin, **M. Madhyastha**, F. Sun, N. J. Yadwadkar, C.-J. Wu. *Composer: A Search Framework for Hybrid Neural Architecture Design.* ICLR 2026

L. Xue, **M. Madhyastha**, R. Burns, M. Lee, M. K. Marina. *Towards Decentralized and Sustainable Foundation Model Training with the Edge.* ACM SIGEnergy Energy Informatics Review 2025

R. Underwood, **M. Madhyastha**, R. Burns, B. Nicolae. *EvoStore: Towards Scalable Storage of Evolving Learning Models.* HPDC 2024

M. Madhyastha, T. Budavári, V. Braverman, J. Vogelstein, R. Burns. *T-Rex: Reformulating Decision Tree Traversal as Hyperrectangle Enclosure.* ICDE 2024

B. Wheatman, **M. Madhyastha**, R. Burns. *Masked Matrix Multiplication for Emergent Sparsity.* arXiv 2024

M. Madhyastha, R. Underwood, R. Burns, B. Nicolae. *DStore: A Lightweight Scalable Learning Model Repository with Fine-Grain Tensor-Level Access.* ICS 2023

R. Underwood, **M. Madhyastha**, R. Burns, B. Nicolae. *Understanding Patterns of Deep Learning Model Evolution in Network Architecture Search.* HiPC 2023

M. Madhyastha, K. Lillaney, J. Browne, J. Vogelstein, R. Burns. *BLOCKSET: Reducing Inference Latency for Tree-Ensemble Deployment.* KDD 2021

M. Madhyastha, G. Li, V. Strnadová-Neeley, J. Browne, J. Vogelstein, R. Burns, C. Priebe. *Geodesic Forests.* KDD 2020

S. Roy, **M. Madhyastha**, S. Lawrence, V. Rajan. *Inferring Concept Prerequisite Relations from Online Educational Resources.* AAAI/IAAI 2019

SKILLS

Inference	vLLM, SGLang, LMCache, kvcached, FlashInfer, CUDA graphs, KV-cache and prefix caching, MLA and hybrid (Mamba) attention, MoE serving
Performance	GPU profiling, CUDA, sparsity, I/O and memory-layout optimization
Infrastructure	Kubernetes GPU scheduling and device plugins, Helm, model distribution, Grafana / Prometheus / Loki
Languages	Python, C++, CUDA, Go, Bash